

Chapter 6

Lecture 2.

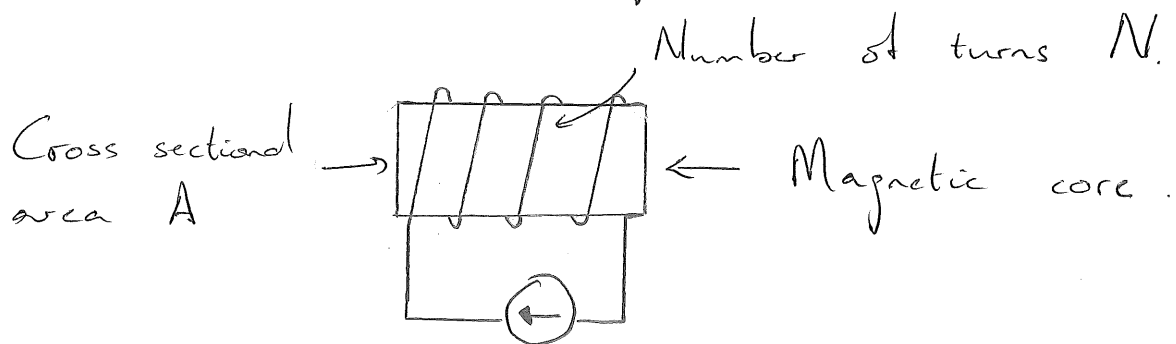
①

* Inductors



→ Passive element.

→ Stores energy in magnetic field.



→ Inductor is a coil of conducting wire around a magnetic core.

$N \Rightarrow$ number of turns

$A \Rightarrow$ cross sectional area $[m^2]$

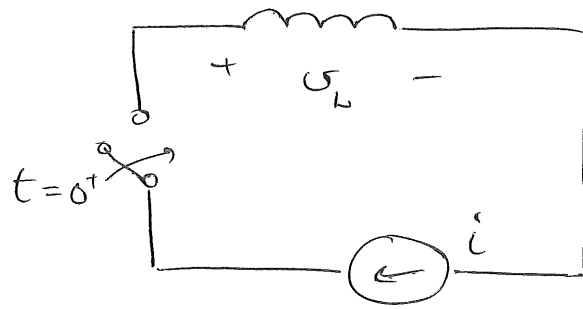
$\mu \Rightarrow$ permeability of core $[H/m]$

$l \Rightarrow$ length of coil.

$$L = \frac{N^2 \mu A}{l} \quad [H] = \text{Henry.}$$

(2)

* What does an inductor do?



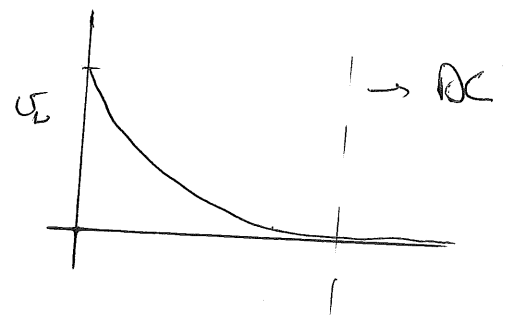
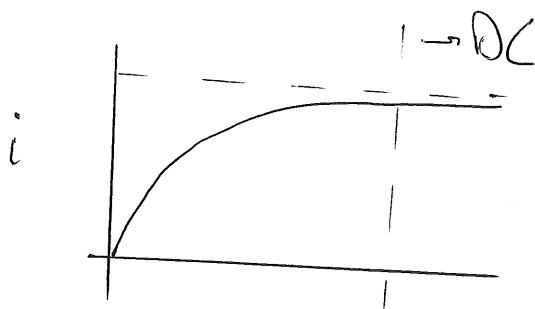
* At $t < 0^+$; there is no magnetic field,

* At $t = 0^+$ switch is closed.

Current flows through inductor and magnetic field builds around inductor.

* As magnetic field grows, it resists flow of current. Voltage is induced in wires.

* When magnetic field is at maximum, no more voltage is induced.



* DC-state
or steady-state.

* Equations

$$\begin{array}{c} i_L \rightarrow \\ \text{---} \\ + \quad V_L \quad - \end{array}$$

$$\rightarrow V_L = L \frac{di_L}{dt}$$

$$\rightarrow \therefore i_L = \frac{1}{L} \int_{t_0}^t V_L dt + i_L(t_0)$$

$$\rightarrow p = V_L i_L = i_L \times L \frac{di_L}{dt}$$

$$\rightarrow W_L = \frac{1}{2} L i_L^2$$

* DC conditions (V and I constant)

* If i_L is constant:

$$V_L = L \frac{di_L}{dt} = 0 \text{ V.}$$

* Inductor is short circuit at DC.

* Series + Parallel

* Same as resistors

$$\rightarrow \text{Series: } L_{eq} = L_1 + L_2 + L_3 \dots$$

$$\rightarrow \text{Parallel: } \frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \dots$$

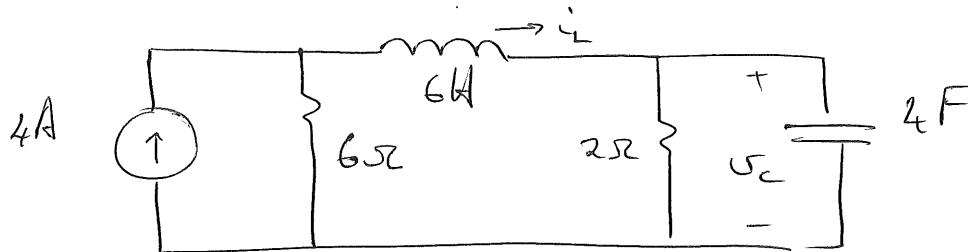
* Examples

④

① $L = 1 \text{ mH}$ $i(t) = 20 \cos(100t) \text{ mA}$.
 $v = ?$ $\omega = ?$

② $L = 2 \text{ H}$ $v(t) = 10(1-t) \text{ V}$
 i at $t = 4 \text{ s}$ if $i(0) = 2 \text{ A}$?

③ $v_c = ?$ $i_L = ?$ $\omega_c = ?$ $\omega_L = ?$



④ $L_{eq} = ?$

